

WESA MWITI

Cyber security Specialist & Backend Software Engineer

<https://mwitiwesa-coke.github.io/Portfolio/>
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Profile

Passionate Cybersecurity professional with foundational knowledge in defensive practices, secure development, threat mitigation, and basic network security to build resilient systems against evolving attacks. Strong backend software engineering foundation for implementing secure APIs, databases, and applications. Experienced in developing protected RESTful APIs for secure data storage, vulnerability assessment, and team leadership. Focused on empowering startups and businesses with efficient, breach-resilient solutions through secure coding, incident response basics and compliance awareness (OWASP, NIST). Skilled in coding, basic threat analysis, backend development, group management, leadership, and team collaboration.

Skills

- Build and maintain secure web applications and RESTful API (authentication, encryption, input validation, OWASP compliance)
- Identify, debug, and mitigate issues including security vulnerabilities, production threats, and basic anomaly detection.
- Design secure database schemas (PostgreSQL, SQLite, MySQL0 with encryption, least privilege access, query security, and basic endpoint security principles)
- Develop protected user interfaces (html5, css3, JavaScript, Bootstrap).
- Integrate frontend with Django backend views and secure REST APIs.
- Basic network security, firewall management basics, monitoring systems, report writing and cybersecurity awareness training.

Cybersecurity Tools & Technologies

- Web/API security testing: Burp suite, OWASP ZAP
- Vulnerability scanning & code analysis: Sync, SonarQube, Nikto
- Network & packet analysis: Wireshark, Nmap (network mapping & basic traffic analysis)
- Secure development practices: OWASP, JWT/OAuth authentication, TLS/HTTPS enforcement.
- Asset Discovery – DNS, Amass Shodan
- Organization Information Harvesting – theHarvester h8mail
- Open-Source Intelligence gathering (OSINT)– Recon-ng Framework, SpiderFoot

Languages & Technologies

- Python – secure scripting, APIs, data pipelines, ML, automation
- JavaScript / TypeScript – secure frontend (React, Angular, Vue)
- Node.js – scalable backend systems, Android integration
- SQL (PostgreSQL, MySQL, SQL Server) – secure relational databases
- NoSQL (MongoDB, Redis, Cassandra) – distributed stores with encryption and access controls

Education

- Cybersecurity – ALX Africa 2025 – Graduated 2026
- Software Engineering – ALX Africa – Specialized in Backend Development – Graduated 2025
- ALX Ventures - Founder – ALX Africa – Graduated 2024
- Professional Foundations – ALX Africa – Graduated 2025
- Fisheries and Aquatic Science, WRITI – Naivasha

Projects

- B-safe (UX Researcher & Developer – Team Elephant) Mobile/web app for anonymous abuse reporting to combat femicide in Kenya. Delivered secure backend with encrypted data handling, privacy controls, basic security audits, and successful deployment.
- Jabali (Offline AI Assistant) (Backend Engineer) Desktop AI featuring offline speech recognition, biometric face recognition login, GUI, and task automation. Implemented secure local storage, authentication, and data protection features.
- PointaFam (Backend Engineer) Platform connecting farmers to consumers/wholesalers, eliminating brokers for better profits. Built secure transaction APIs, user data safeguards, and basic access controls.
- TicketX (Backend Engineer & Cybersecurity Contributor) Event ticketing platform with secure authentication, payment integration, real-time booking, and basic monitoring. Applied secure development and cybersecurity practices to prevent breaches and ensure data integrity.